



Technical Service Bulletin: TSB230177

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Loose or Worn Accessory Drive Assembly and Damage Mitigation Recommendations

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QSK23 CM500
- QSK23 CM2250 K109
- QSK23 CM2350 K127

Issue Summary

Symptom:

- Excessive noise from front side of engine
- Seized and overheated accessory drive
- Worn or malfunctioning accessory drive bearings

Root Cause:

- Insufficient lubrication
- Improper belt tension

Verification

Seized or malfunctioning accessory drive bearings resulting in damaged or malfunctioning adaptor assemblies. See Figure 1 and Figure 2 below.



Figure 1, Damaged, Scuffed Accessory Drive Shaft.



Figure 2, Damaged, Scuffed Accessory Drive Body.

Resolution

The new accessory drive **must** be lubricated (greased) manually before installation. As a malfunction preventative measure/mitigation, lubricate (grease) in service drives at 5000 hour intervals beginning with the next appropriate service interval. See Service Instructions section.

Service Instructions

Place the new accessory drive in a vice in a horizontal position. The accessory drive axis (Figure 3, Item 2) should be parallel to floor, with the inlet hole (Figure 3, Item 1) facing the top.

For In- Service Drives:

- Reference corresponding Service Manual. Reference Section 009-009 in Section 9. See equipment manufacturer service information to remove and install the accessory drive and related components.

Applying Grease in Both Cases:

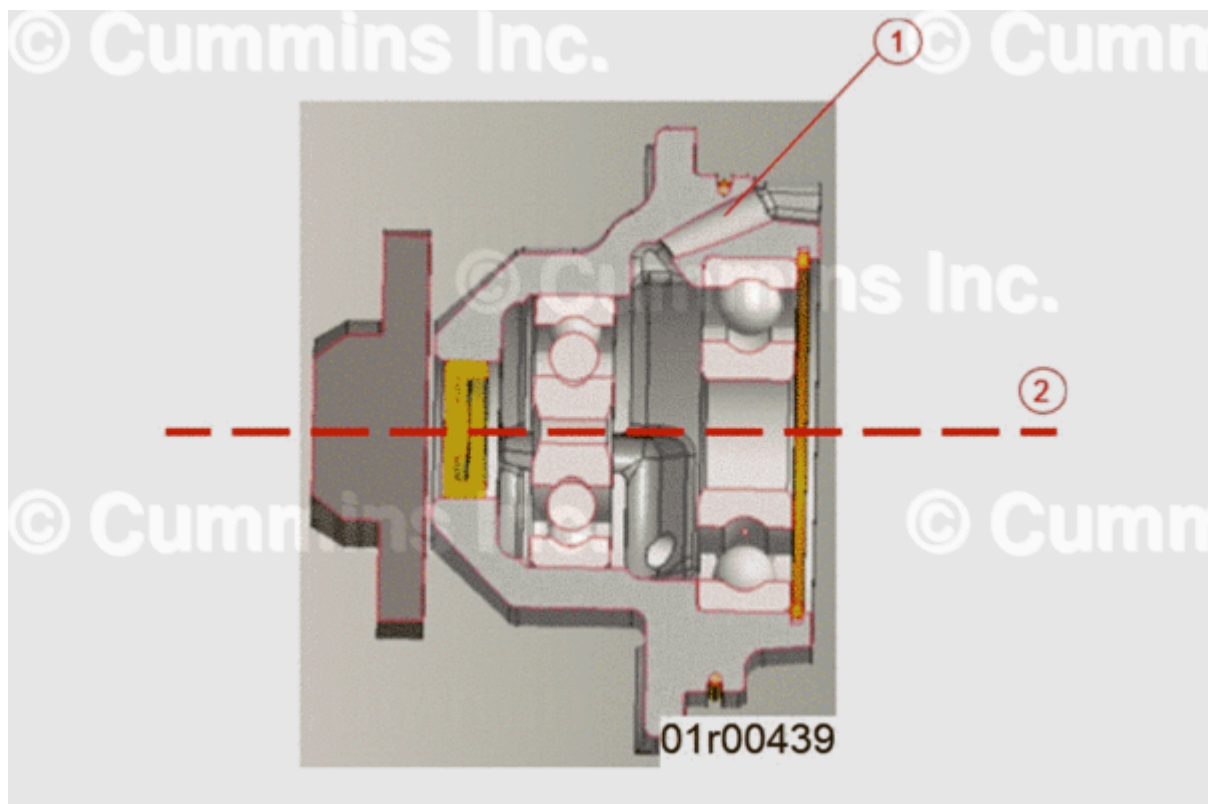


Figure 3, CAD Model of Physical Horizontally Mounted Assessor Drive.

Attach no. 2 general purpose grease onto a grease gun and fit the tip of the nozzle of the grease gun into the inlet hole positioned at the top (Figure 3, Item 2). See Figure 4 for reference.



Figure 4, Grease Gun Fitted to the Top Inlet Hole of Accessory Drive.

Direct inject the grease into the inlet hole of the accessory drive until grease can be seen coming out from the outlet hole or from the bottom side of the gear (whichever is seen first). See Figure 5 for reference of grease coming out from bottom side of gear.



Figure 5, Accessory Drive Injected with Grease with Grease Visible at Underside of Gear.

Important: Upon installation of the accessory drive, follow all related instructions and verify that belts are properly tensioned within recommended specifications (for tensions set manually). For single v-belt refrigerant compressor where an OEM tension specification is **not** available, a tension of 75 to 80 ft-lb may be appropriate.

Document History

Date	Details
2023-9-12	Module Created
2023-12-12	Updated title. Added QSK23 CM2250 K109. Updated Symptom, Root Cause, Verification, and Resolution.

Last Modified: 13-Dec-2023